

MT-104

Linear Algebra

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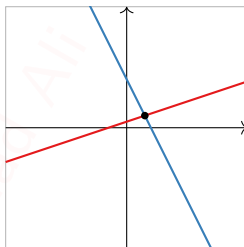
Section 1.3

Vector Equations

Motivation

We want to think about the *algebra* in linear algebra (systems of equations and their solution sets) in terms of *geometry* (points, lines, planes, etc).

$$\begin{aligned}x - 3y &= -3 \\ 2x + y &= 8\end{aligned}$$



This will give us better insight into the properties of systems of equations and their solution sets.

To do this, we need to introduce n -dimensional space $\mathbf{R}^n$, and **vectors** inside it.

Line, Plane, Space, ...

Recall that $\mathbf{R}$ denotes the collection of all real numbers, i.e. the number line.

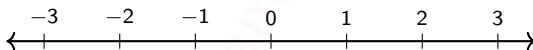
Definition

Let n be a positive whole number. We define

$$\mathbf{R}^n = \text{all ordered } n\text{-tuples of real numbers } (x_1, x_2, x_3, \dots, x_n).$$

Example

When $n = 1$, we just get $\mathbf{R}$ back: $\mathbf{R}^1 = \mathbf{R}$. Geometrically, this is the *number line*.

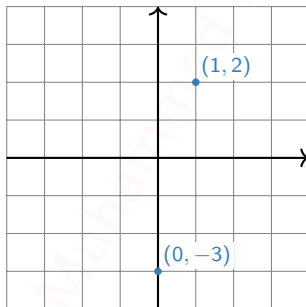


Line, Plane, Space, ...

Continued

Example

When $n = 2$, we can think of $\mathbf{R}^2$ as the *plane*. This is because every point on the plane can be represented by an ordered pair of real numbers, namely, its x - and y -coordinates.



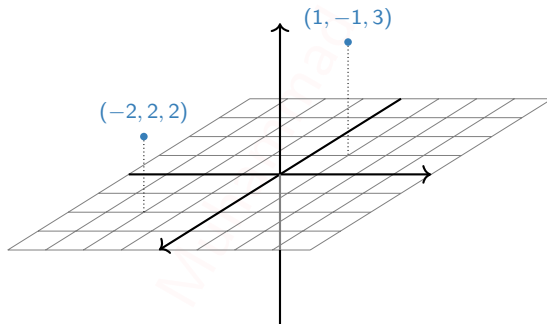
We can use the elements of $\mathbf{R}^2$ to *label* points on the plane, but $\mathbf{R}^2$ is not defined to be the plane!

Line, Plane, Space, ...

Continued

Example

When $n = 3$, we can think of $\mathbf{R}^3$ as the *space* we (appear to) live in. This is because every point in space can be represented by an ordered triple of real numbers, namely, its x -, y -, and z -coordinates.



Again, we can use the elements of $\mathbf{R}^3$ to *label* points in space, but $\mathbf{R}^3$ is not defined to be space!

Line, Plane, Space, ...

Continued

So what is $\mathbf{R}^4$? or $\mathbf{R}^5$? or $\mathbf{R}^n$?

...go back to the *definition*: ordered n -tuples of real numbers

$$(x_1, x_2, x_3, \dots, x_n).$$

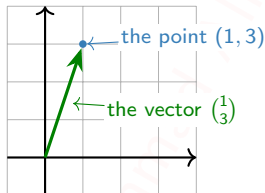
They're still "geometric" spaces, in the sense that our intuition for $\mathbf{R}^2$ and $\mathbf{R}^3$ sometimes extends to $\mathbf{R}^n$, but they're harder to visualize.

We'll make definitions and state theorems that apply to any $\mathbf{R}^n$, but we'll only draw pictures for $\mathbf{R}^2$ and $\mathbf{R}^3$.

Vectors

In the previous slides, we were thinking of elements of $\mathbf{R}^n$ as **points**: in the line, plane, space, etc.

We can also think of them as **vectors**: arrows with a given length and direction.



So the vector points *horizontally* in the amount of its x -coordinate, and *vertically* in the amount of its y -coordinate.

When we think of an element of $\mathbf{R}^n$ as a vector, we write it as a matrix with n rows and one column:

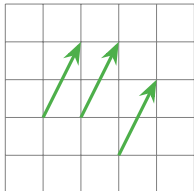
$$v = \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}.$$

We'll see why this is useful later.

Points and Vectors

So what is the difference between a point and a vector?

A vector need not start at the origin: *it can be located anywhere!* In other words, an arrow is determined by its length and its direction, not by its location.



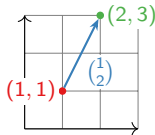
These arrows all represent the vector $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$.

However, unless otherwise specified, we'll assume a vector starts at the origin: we'll usually be sloppy and identify the vector $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ with the point $(1, 2)$.

This makes sense in the real world: many physical quantities, such as velocity, are represented as vectors. But it makes more sense to think of the velocity of a car as being located at the car.

Another way to think about it: a vector is a *difference* between two points, or the arrow from one point to another.

For instance, $\begin{pmatrix} 1 \\ 2 \end{pmatrix}$ is the arrow from $(1, 1)$ to $(2, 3)$.



Vector Algebra

Definition

- We can add two vectors together:

$$\begin{pmatrix} a \\ b \\ c \end{pmatrix} + \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} a+x \\ b+y \\ c+z \end{pmatrix}.$$

- We can multiply, or **scale**, a vector by a real number c :

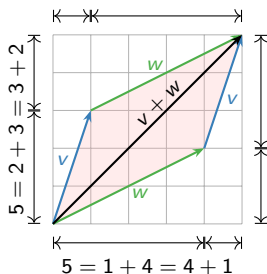
$$c \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} c \cdot x \\ c \cdot y \\ c \cdot z \end{pmatrix}.$$

We call c a **scalar** to distinguish it from a vector. If v is a vector and c is a scalar, cv is called a **scalar multiple** of v .

(And likewise for vectors of length n .) For instance,

$$\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} + \begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix} = \begin{pmatrix} 5 \\ 7 \\ 9 \end{pmatrix} \quad \text{and} \quad -2 \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = \begin{pmatrix} -2 \\ -4 \\ -6 \end{pmatrix}.$$

Vector Addition and Subtraction: Geometry



The parallelogram law for vector addition

Geometrically, the sum of two vectors v, w is obtained as follows: place the tail of w at the head of v . Then $v + w$ is the vector whose tail is the tail of v and whose head is the head of w . Doing this both ways creates a **parallelogram**. For example,

$$\begin{pmatrix} 1 \\ 3 \end{pmatrix} + \begin{pmatrix} 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 5 \\ 5 \end{pmatrix}.$$

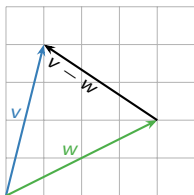
Why? The width of $v + w$ is the sum of the widths, and likewise with the heights.

Vector subtraction

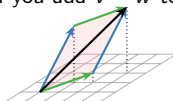
Geometrically, the difference of two vectors v, w is obtained as follows: place the tail of v and w at the same point. Then $v - w$ is the vector from the head of v to the head of w . For example,

$$\begin{pmatrix} 1 \\ 4 \end{pmatrix} - \begin{pmatrix} 4 \\ 2 \end{pmatrix} = \begin{pmatrix} -3 \\ 2 \end{pmatrix}.$$

Why? If you add $v - w$ to w , you get v .



This works in higher dimensions too!

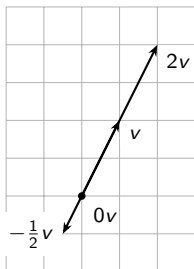


Scalar Multiplication: Geometry

Scalar multiples of a vector

These have the same *direction* but a different *length*.

Some multiples of v .



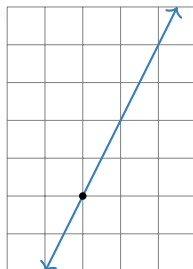
$$v = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

$$2v = \begin{pmatrix} 2 \\ 4 \end{pmatrix}$$

$$-\frac{1}{2}v = \begin{pmatrix} -\frac{1}{2} \\ -1 \end{pmatrix}$$

$$0v = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

All multiples of v .



So the scalar multiples of v form a *line*.

Linear Combinations

We can add and scalar multiply in the same equation:

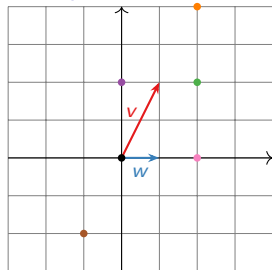
$$w = c_1 v_1 + c_2 v_2 + \cdots + c_p v_p$$

where $c_1, c_2, \dots, c_p$ are scalars, $v_1, v_2, \dots, v_p$ are vectors in $\mathbf{R}^n$, and w is a vector in $\mathbf{R}^n$.

Definition

We call w a **linear combination** of the vectors $v_1, v_2, \dots, v_p$. The scalars $c_1, c_2, \dots, c_p$ are called the **weights** or **coefficients**.

Example



Let $v = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ and $w = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$.

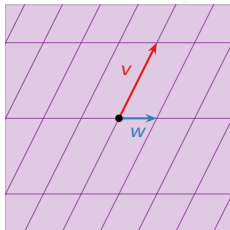
What are some linear combinations of v and w ?

- ▶ $v + w$
- ▶ $v - w$
- ▶ $2v + 0w$
- ▶ $2w$
- ▶ $-v$

Poll

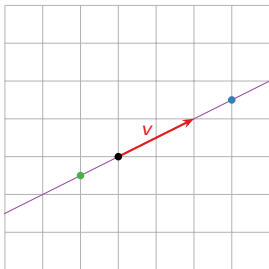
Is there any vector in $\mathbf{R}^2$ that is *not* a linear combination of v and w ?

No: in fact, every vector in $\mathbf{R}^2$ is a combination of v and w .



(The purple lines are to help measure *how much* of v and w you need to get to a given point.)

More Examples



What are some linear combinations of $v = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$?

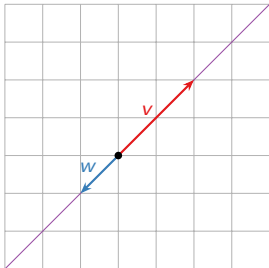
▶ $\frac{3}{2}v$

▶ $-\frac{1}{2}v$

▶ ...

What are *all* linear combinations of v ?

All vectors cv for c a real number. I.e., all *scalar multiples* of v . These form a *line*.



Question

What are all linear combinations of

$$v = \begin{pmatrix} 2 \\ 2 \end{pmatrix} \quad \text{and} \quad w = \begin{pmatrix} -1 \\ -1 \end{pmatrix}?$$

Answer: The line which contains both vectors.

What's different about this example and the one on the poll?

Systems of Linear Equations

Question

Is $\begin{pmatrix} 8 \\ 16 \\ 3 \end{pmatrix}$ a linear combination of $\begin{pmatrix} 1 \\ 2 \\ 6 \end{pmatrix}$ and $\begin{pmatrix} -1 \\ -2 \\ -1 \end{pmatrix}$?

This means: can we solve the equation

$$x \begin{pmatrix} 1 \\ 2 \\ 6 \end{pmatrix} + y \begin{pmatrix} -1 \\ -2 \\ -1 \end{pmatrix} = \begin{pmatrix} 8 \\ 16 \\ 3 \end{pmatrix}$$

where x and y are the unknowns (the coefficients)? Rewrite:

$$\begin{pmatrix} x \\ 2x \\ 6x \end{pmatrix} + \begin{pmatrix} -y \\ -2y \\ -y \end{pmatrix} = \begin{pmatrix} 8 \\ 16 \\ 3 \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} x - y \\ 2x - 2y \\ 6x - y \end{pmatrix} = \begin{pmatrix} 8 \\ 16 \\ 3 \end{pmatrix}.$$

This is just a system of linear equations:

$$\begin{aligned} x - y &= 8 \\ 2x - 2y &= 16 \\ 6x - y &= 3. \end{aligned}$$

Systems of Linear Equations

Continued

$$x - y = 8$$

$$2x - 2y = 16$$

$$6x - y = 3$$

matrix form
~~~~~>

$$\left( \begin{array}{cc|c} 1 & -1 & 8 \\ 2 & -2 & 16 \\ 6 & -1 & 3 \end{array} \right)$$

row reduce  
~~~~~>

$$\left(\begin{array}{cc|c} 1 & 0 & -1 \\ 0 & 1 & -9 \\ 0 & 0 & 0 \end{array} \right)$$

solution
~~~~~>

$$x = -1$$

$$y = -9$$

Conclusion:

$$-\begin{pmatrix} 1 \\ 2 \\ 6 \end{pmatrix} - 9\begin{pmatrix} -1 \\ -2 \\ -1 \end{pmatrix} = \begin{pmatrix} 8 \\ 16 \\ 3 \end{pmatrix}$$

What is the relationship between the original vectors and the matrix form of the linear equation? They have the same columns!

**Shortcut:** You can make the augmented matrix without writing down the system of linear equations first.

## Vector Equations and Linear Equations

### Summary

The vector equation

$$x_1 v_1 + x_2 v_2 + \cdots + x_p v_p = b,$$

where  $v_1, v_2, \dots, v_p, b$  are vectors in  $\mathbf{R}^n$  and  $x_1, x_2, \dots, x_p$  are scalars, has the same solution set as the linear system with augmented matrix

$$\left( \begin{array}{c|c|c|c|c} | & | & & | & | \\ v_1 & v_2 & \cdots & v_p & b \\ | & | & & | & | \end{array} \right),$$

where the  $v_i$ 's and  $b$  are the columns of the matrix.

So we now have (at least) *two* equivalent ways of thinking about linear systems of equations:

1. Augmented matrices.
2. Linear combinations of vectors (vector equations).

The last one is more geometric in nature.

# Span

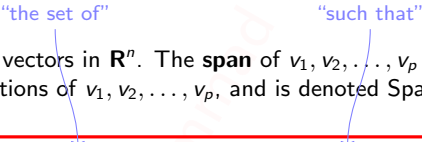
It is important to know what are *all* linear combinations of a set of vectors  $v_1, v_2, \dots, v_p$  in  $\mathbf{R}^n$ : it's exactly the collection of all  $b$  in  $\mathbf{R}^n$  such that the vector equation (in the unknowns  $x_1, x_2, \dots, x_p$ )

$$x_1 v_1 + x_2 v_2 + \dots + x_p v_p = b$$

has a solution (i.e., is consistent).

## Definition

Let  $v_1, v_2, \dots, v_p$  be vectors in  $\mathbf{R}^n$ . The **span** of  $v_1, v_2, \dots, v_p$  is the collection of all linear combinations of  $v_1, v_2, \dots, v_p$ , and is denoted  $\text{Span}\{v_1, v_2, \dots, v_p\}$ . In symbols:


$$\text{Span}\{v_1, v_2, \dots, v_p\} = \{x_1 v_1 + x_2 v_2 + \dots + x_p v_p \mid x_1, x_2, \dots, x_p \text{ in } \mathbf{R}\}.$$

**Synonyms:**  $\text{Span}\{v_1, v_2, \dots, v_p\}$  is the subset **spanned by** or **generated by**  $v_1, v_2, \dots, v_p$ .

This is the first of several definitions in this class that you simply **must learn**. I will give you other ways to think about Span, and ways to draw pictures, but *this is the definition*. Having a vague idea what Span means will not help you solve any exam problems!

# Span

## Continued

Now we have several equivalent ways of making the same statement:

1. A vector  $b$  is in the span of  $v_1, v_2, \dots, v_p$ .
2. The linear system with augmented matrix

$$\left( \begin{array}{c|c|c|c|c} | & | & & | & | \\ v_1 & v_2 & \cdots & v_p & b \\ | & | & & | & | \end{array} \right)$$

is consistent.

3. The vector equation

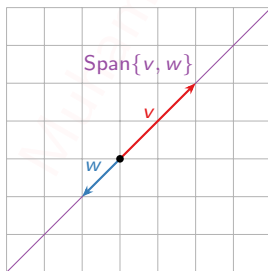
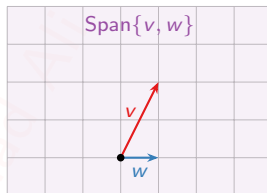
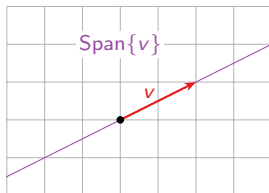
$$x_1 v_1 + x_2 v_2 + \cdots + x_p v_p = b$$

has a solution.

**Note:** **equivalent** means that, for any given list of vectors  $v_1, v_2, \dots, v_p, b$ , *either* all three statements are true, *or* all three statements are false.

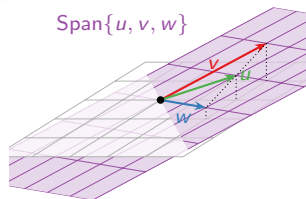
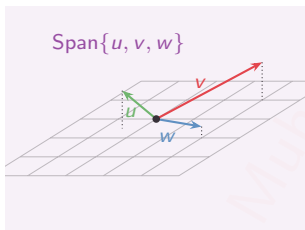
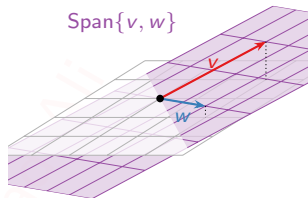
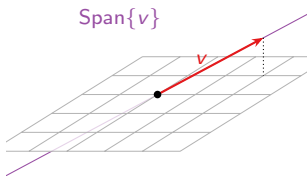
## Pictures of Span

Drawing a picture of  $\text{Span}\{v_1, v_2, \dots, v_p\}$  is the same as drawing a picture of all linear combinations of  $v_1, v_2, \dots, v_p$ .



# Pictures of Span

In  $\mathbb{R}^3$



Poll

How many vectors are in  $\text{Span} \left\{ \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \right\}$ ?

- A. Zero
- B. One
- C. Infinity

In general, it appears that  $\text{Span}\{v_1, v_2, \dots, v_p\}$  is the smallest “linear space” (line, plane, etc.) containing the origin and all of the vectors  $v_1, v_2, \dots, v_p$ .

We will make this precise later.

## Matrix $\times$ Vector

the first number is  
the number of rows

the second number is  
the number of columns

Let  $A$  be an  $m \times n$  matrix

$$A = \begin{pmatrix} | & | & \cdots & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & \cdots & | \end{pmatrix} \quad \text{with columns } v_1, v_2, \dots, v_n$$

### Definition

The **product** of  $A$  with a vector  $x$  in  $\mathbf{R}^n$  is the linear combination

$$Ax = \begin{pmatrix} | & | & \cdots & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & \cdots & | \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \stackrel{\text{def}}{=} x_1 v_1 + x_2 v_2 + \cdots + x_n v_n.$$

Annotations:  
- Blue arrow from "this means the equality is a definition" to the definition symbol  $\stackrel{\text{def}}{=}$ .  
- Red arrow from "these must be equal" pointing to the  $v_n$  and  $x_n$  terms in the matrix product.

The output is a vector in  $\mathbf{R}^m$ .

Note that the number of **columns** of  $A$  has to equal the number of **rows** of  $x$ .

### Example

$$\begin{pmatrix} 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix} = 1 \begin{pmatrix} 4 \\ 7 \end{pmatrix} + 2 \begin{pmatrix} 5 \\ 8 \end{pmatrix} + 3 \begin{pmatrix} 6 \\ 9 \end{pmatrix} = \begin{pmatrix} 32 \\ 50 \end{pmatrix}.$$



# Matrix Equations

An example

## Question

Let  $v_1, v_2, v_3$  be vectors in  $\mathbf{R}^3$ . How can you write the vector equation

$$2v_1 + 3v_2 - 4v_3 = \begin{pmatrix} 7 \\ 2 \\ 1 \end{pmatrix}$$

in terms of matrix multiplication?

**Answer:** Let  $A$  be the matrix with columns  $v_1, v_2, v_3$ , and let  $x$  be the vector with entries  $2, 3, -4$ . Then

$$Ax = \begin{pmatrix} | & | & | \\ v_1 & v_2 & v_3 \\ | & | & | \end{pmatrix} \begin{pmatrix} 2 \\ 3 \\ -4 \end{pmatrix} = 2v_1 + 3v_2 - 4v_3,$$

so the vector equation is equivalent to the matrix equation

$$Ax = \begin{pmatrix} 7 \\ 2 \\ 1 \end{pmatrix}.$$

# Matrix Equations

In general

Let  $v_1, v_2, \dots, v_n$ , and  $b$  be vectors in  $\mathbf{R}^m$ . Consider the vector equation

$$x_1 v_1 + x_2 v_2 + \dots + x_n v_n = b.$$

It is equivalent to the **matrix equation**

$$Ax = b$$

where

$$A = \begin{pmatrix} | & | & \cdots & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & \cdots & | \end{pmatrix} \quad \text{and} \quad x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}.$$

Conversely, if  $A$  is any  $m \times n$  matrix, then

$$Ax = b \quad \text{is equivalent to the} \quad x_1 v_1 + x_2 v_2 + \dots + x_n v_n = b$$

vector equation

where  $v_1, \dots, v_n$  are the columns of  $A$ , and  $x_1, \dots, x_n$  are the entries of  $x$ .

## Linear Systems, Vector Equations, Matrix Equations, ...

We now have *four* equivalent ways of writing (and thinking about) linear systems:

1. As a system of equations:

$$2x_1 + 3x_2 = 7$$

$$x_1 - x_2 = 5$$

2. As an augmented matrix:

$$\left( \begin{array}{cc|c} 2 & 3 & 7 \\ 1 & -1 & 5 \end{array} \right)$$

3. As a vector equation ( $x_1 v_1 + \cdots + x_n v_n = b$ ):

$$x_1 \begin{pmatrix} 2 \\ 1 \end{pmatrix} + x_2 \begin{pmatrix} 3 \\ -1 \end{pmatrix} = \begin{pmatrix} 7 \\ 5 \end{pmatrix}$$

4. As a matrix equation ( $Ax = b$ ):

$$\begin{pmatrix} 2 & 3 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 7 \\ 5 \end{pmatrix}$$

We will move back and forth freely between these over and over again, for the rest of the semester. Get comfortable with them now!

In particular, *all four have the same solution set.*

## Matrix $\times$ Vector

Another way

### Definition

A **row vector** is a matrix with one row. The product of a row vector of length  $n$  and a (column) vector of length  $n$  is

$$(a_1 \quad \cdots \quad a_n) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \stackrel{\text{def}}{=} a_1 x_1 + \cdots + a_n x_n.$$

This is a scalar.

If  $A$  is an  $m \times n$  matrix with rows  $r_1, r_2, \dots, r_m$ , and  $x$  is a vector in  $\mathbf{R}^n$ , then

$$Ax = \begin{pmatrix} -r_1- \\ -r_2- \\ \vdots \\ -r_m- \end{pmatrix} x = \begin{pmatrix} r_1 x \\ r_2 x \\ \vdots \\ r_m x \end{pmatrix}$$

This is a vector in  $\mathbf{R}^m$  (again).

## Spans and Solutions to Equations

Let  $A$  be a matrix with columns  $v_1, v_2, \dots, v_n$ :

$$A = \begin{pmatrix} | & | & \cdots & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & \cdots & | \end{pmatrix}$$

Very Important Fact That Will Appear on Every Midterm and the Final

$Ax = b$  has a solution

$$\iff \text{there exist } x_1, \dots, x_n \text{ such that } A \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = b$$

"if and only if"

$$\iff \text{there exist } x_1, \dots, x_n \text{ such that } x_1 v_1 + \cdots + x_n v_n = b$$

$$\iff b \text{ is a linear combination of } v_1, \dots, v_n$$

$$\iff b \text{ is in the span of the columns of } A.$$

The last condition is geometric.

# Spans and Solutions to Equations

## Example

### Question

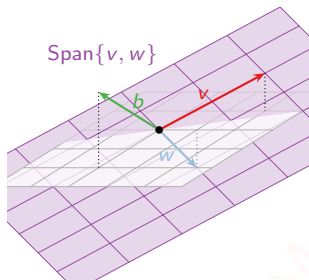
Let  $A = \begin{pmatrix} 2 & 1 \\ -1 & 0 \\ 1 & -1 \end{pmatrix}$ . Does the equation  $Ax = \begin{pmatrix} 0 \\ 2 \\ 2 \end{pmatrix}$  have a solution?

Columns of  $A$ :

$$v = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \quad w = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

Output vector:

$$b = \begin{pmatrix} 0 \\ 2 \\ 2 \end{pmatrix}$$



Is  $b$  contained in the span of the columns of  $A$ ? It sure doesn't look like it.

**Conclusion:**  $Ax = b$  is *inconsistent*.

# Spans and Solutions to Equations

Example, continued

## Question

Let  $A = \begin{pmatrix} 2 & 1 \\ -1 & 0 \\ 1 & -1 \end{pmatrix}$ . Does the equation  $Ax = \begin{pmatrix} 0 \\ 2 \\ 2 \end{pmatrix}$  have a solution?

**Answer:** Let's check by solving the matrix equation using row reduction.

The first step is to put the system into an augmented matrix.

$$\left( \begin{array}{cc|c} 2 & 1 & 0 \\ -1 & 0 & 2 \\ 1 & -1 & 2 \end{array} \right) \xrightarrow{\text{row reduce}} \left( \begin{array}{cc|c} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right)$$

The last equation is  $0 = 1$ , so the system is *inconsistent*.

In other words, the matrix equation

$$\begin{pmatrix} 2 & 1 \\ -1 & 0 \\ 1 & -1 \end{pmatrix} x = \begin{pmatrix} 0 \\ 2 \\ 2 \end{pmatrix}$$

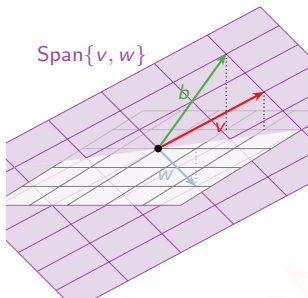
has no solution, as the picture shows.

# Spans and Solutions to Equations

## Example

### Question

Let  $A = \begin{pmatrix} 2 & 1 \\ -1 & 0 \\ 1 & -1 \end{pmatrix}$ . Does the equation  $Ax = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$  have a solution?



Columns of  $A$ :

$$v = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \quad w = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

Solution vector:

$$b = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$$

Is  $b$  contained in the span of the columns of  $A$ ? It looks like it: in fact,

$$b = 1v + (-1)w \implies x = \begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$



# Spans and Solutions to Equations

Example, continued

## Question

Let  $A = \begin{pmatrix} 2 & 1 \\ -1 & 0 \\ 1 & -1 \end{pmatrix}$ . Does the equation  $Ax = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$  have a solution?

**Answer:** Let's do this systematically using row reduction.

$$\left( \begin{array}{cc|c} 2 & 1 & 1 \\ -1 & 0 & -1 \\ 1 & -1 & 2 \end{array} \right) \xrightarrow{\text{row reduce}} \left( \begin{array}{cc|c} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{array} \right)$$

This gives us

$$x = 1 \quad y = -1.$$

This is consistent with the picture on the previous slide:

$$1 \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} - 1 \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} \quad \text{or} \quad A \begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}.$$

## Poll

Which of the following true statements can be checked by eyeballing them, *without* row reduction?

- A.  $\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$  is in the span of  $\begin{pmatrix} 3 \\ 3 \\ 4 \end{pmatrix}$ ,  $\begin{pmatrix} 0 \\ 10 \\ 20 \end{pmatrix}$ ,  $\begin{pmatrix} 0 \\ -1 \\ -2 \end{pmatrix}$ .
- B.  $\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$  is in the span of  $\begin{pmatrix} 3 \\ 3 \\ 4 \end{pmatrix}$ ,  $\begin{pmatrix} 0 \\ 5 \\ 7 \end{pmatrix}$ ,  $\begin{pmatrix} 0 \\ 6 \\ 8 \end{pmatrix}$ .
- C.  $\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$  is in the span of  $\begin{pmatrix} 3 \\ 3 \\ 4 \end{pmatrix}$ ,  $\begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$ ,  $\begin{pmatrix} 0 \\ 0 \\ \sqrt{2} \end{pmatrix}$ .
- D.  $\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$  is in the span of  $\begin{pmatrix} 5 \\ 7 \\ 0 \end{pmatrix}$ ,  $\begin{pmatrix} 6 \\ 8 \\ 0 \end{pmatrix}$ ,  $\begin{pmatrix} 3 \\ 3 \\ 4 \end{pmatrix}$ .

## When Solutions Always Exist

Here are criteria for a linear system to always have a solution.

### Theorem

Let  $A$  be an  $m \times n$  (non-augmented) matrix. The following are equivalent

1.  $Ax = b$  has a solution for all  $b$  in  $\mathbf{R}^m$ .
2. The span of the columns of  $A$  is all of  $\mathbf{R}^m$ .
3.  $A$  has a pivot in each row.

recall that this means  
that for given  $A$ , either they're  
all true, or they're all false

Why is (1) the same as (2)? This was the **Very Important** box from before.

Why is (1) the same as (3)? If  $A$  has a pivot in each row then its reduced row echelon form looks like this:

$$\begin{pmatrix} 1 & 0 & \star & 0 & \star \\ 0 & 1 & \star & 0 & \star \\ 0 & 0 & 0 & 1 & \star \end{pmatrix} \quad \text{and } (A | b) \text{ reduces to this: } \begin{pmatrix} 1 & 0 & \star & 0 & \star & | & \star \\ 0 & 1 & \star & 0 & \star & | & \star \\ 0 & 0 & 0 & 1 & \star & | & \star \end{pmatrix}.$$

There's no  $b$  that makes it inconsistent, so there's always a solution. If  $A$  doesn't have a pivot in each row, then its reduced form looks like this:

$$\begin{pmatrix} 1 & 0 & \star & 0 & \star \\ 0 & 1 & \star & 0 & \star \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad \begin{array}{l} \text{and this can be} \\ \text{made} \\ \text{inconsistent:} \end{array} \quad \begin{pmatrix} 1 & 0 & \star & 0 & \star & | & 0 \\ 0 & 1 & \star & 0 & \star & | & 0 \\ 0 & 0 & 0 & 0 & 0 & | & 16 \end{pmatrix}.$$

## Properties of the Matrix–Vector Product

Let  $c$  be a scalar,  $u, v$  be vectors, and  $A$  a matrix.

►  $A(u + v) = Au + Av$

►  $A(cv) = cAv$

See Lay, §1.4, Theorem 5.

For instance,  $A(3u - 7v) = 3Au - 7Av$ .

**Consequence:** If  $u$  and  $v$  are solutions to  $Ax = 0$ , then so is every vector in  $\text{Span}\{u, v\}$ . Why?

$$\begin{cases} Au = 0 \\ Av = 0 \end{cases} \implies A(xu + yv) = xAu + yAv = x0 + y0 = 0.$$

(Here  $0$  means the zero vector.)

**Important**

The set of solutions to  $Ax = 0$  is a span.

## 1.4 EXERCISES

Compute the products in Exercises 1–4 using (a) the definition, as in Example 1, and (b) the row–vector rule for computing  $Ax$ . If a product is undefined, explain why.

$$1. \begin{bmatrix} -4 & 2 \\ 1 & 6 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ -2 \\ 7 \end{bmatrix} \quad 2. \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} \begin{bmatrix} 5 \\ -1 \end{bmatrix}$$

$$3. \begin{bmatrix} 1 & 2 \\ -3 & 1 \\ 1 & 6 \end{bmatrix} \begin{bmatrix} -2 \\ 3 \end{bmatrix} \quad 4. \begin{bmatrix} 1 & 3 & -4 \\ 3 & 2 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$$

In Exercises 5–8, use the definition of  $Ax$  to write the matrix equation as a vector equation, or vice versa.

$$5. \begin{bmatrix} 1 & 2 & -3 & 1 \\ -2 & -3 & 1 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ -1 \\ 1 \\ -1 \end{bmatrix} = \begin{bmatrix} -4 \\ 1 \end{bmatrix}$$

$$6. \begin{bmatrix} 2 & -3 \\ 3 & 2 \\ 8 & -5 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} -3 \\ 5 \end{bmatrix} = \begin{bmatrix} -21 \\ 1 \\ -49 \\ 11 \end{bmatrix}$$

$$7. x_1 \begin{bmatrix} 4 \\ -1 \\ 7 \\ -4 \end{bmatrix} + x_2 \begin{bmatrix} -5 \\ 3 \\ 0 \\ 1 \end{bmatrix} + x_3 \begin{bmatrix} 7 \\ -8 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 6 \\ -8 \\ 0 \\ -7 \end{bmatrix}$$

$$8. z_1 \begin{bmatrix} 2 \\ -4 \end{bmatrix} + z_2 \begin{bmatrix} -1 \\ 5 \end{bmatrix} + z_3 \begin{bmatrix} -4 \\ 3 \end{bmatrix} + z_4 \begin{bmatrix} 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 12 \end{bmatrix}$$

In Exercises 9 and 10, write the system first as a vector equation and then as a matrix equation.

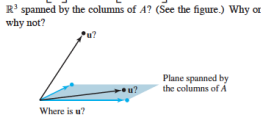
$$9. \begin{aligned} 5x_1 + x_2 - 3x_3 &= 8 \\ 2x_2 + 4x_3 &= 0 \end{aligned} \quad 10. \begin{aligned} 4x_1 - x_2 &= 8 \\ 5x_1 + 3x_2 &= 2 \\ 3x_1 - x_2 &= 1 \end{aligned}$$

Given  $A$  and  $b$  in Exercises 11 and 12, write the augmented matrix for the linear system that corresponds to the matrix equation  $Ax = b$ . Then solve the system and write the solution as a vector.

$$11. A = \begin{bmatrix} 1 & 3 & -4 \\ 1 & 5 & 2 \\ -3 & -7 & 6 \end{bmatrix}, b = \begin{bmatrix} -2 \\ 4 \\ 12 \end{bmatrix}$$

$$12. A = \begin{bmatrix} 1 & 2 & -1 \\ -3 & -4 & 2 \\ 5 & 2 & 3 \end{bmatrix}, b = \begin{bmatrix} 1 \\ 2 \\ -3 \end{bmatrix}$$

$$13. \text{ Let } u = \begin{bmatrix} 0 \\ 4 \\ 4 \end{bmatrix} \text{ and } A = \begin{bmatrix} 3 & -5 \\ -2 & 6 \\ 1 & 1 \end{bmatrix}. \text{ Is } u \text{ in the plane in } \mathbb{R}^3 \text{ spanned by the columns of } A? \text{ (See the figure.) Why or why not?}$$



$$14. \text{ Let } u = \begin{bmatrix} 4 \\ -1 \\ 4 \end{bmatrix} \text{ and } A = \begin{bmatrix} 2 & 5 & -1 \\ 0 & 1 & -1 \\ 1 & 2 & 0 \end{bmatrix}. \text{ Is } u \text{ in the subset of } \mathbb{R}^3 \text{ spanned by the columns of } A? \text{ Why or why not?}$$

15. Let  $A = \begin{bmatrix} 3 & -1 \\ -9 & 3 \end{bmatrix}$  and  $b = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}$ . Show that the equation  $Ax = b$  does not have a solution for all possible  $b$ , and describe the set of all  $b$  for which  $Ax = b$  does have a solution.

16. Repeat the requests from Exercise 15 with

$$A = \begin{bmatrix} 1 & -2 & -1 \\ -2 & 2 & 0 \\ 4 & -1 & 3 \end{bmatrix}, \quad \text{and} \quad b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}.$$

Exercises 17–20 refer to the matrices  $A$  and  $B$  below. Make appropriate calculations that justify your answers and mention an appropriate theorem.

$$A = \begin{bmatrix} 1 & 3 & 0 & 3 \\ -1 & -1 & -1 & 1 \\ 0 & -4 & 2 & -8 \\ 2 & 0 & 3 & -1 \end{bmatrix} \quad B = \begin{bmatrix} 1 & 4 & 1 & 2 \\ 0 & 1 & 3 & -4 \\ 0 & 2 & 6 & 7 \\ 2 & 9 & 5 & -7 \end{bmatrix}$$

17. How many rows of  $A$  contain a pivot position? Does the equation  $Ax = b$  have a solution for each  $b$  in  $\mathbb{R}^4$ ?
18. Can every vector in  $\mathbb{R}^4$  be written as a linear combination of the columns of the matrix  $B$  above? Do the columns of  $B$  span  $\mathbb{R}^4$ ?
19. Can each vector in  $\mathbb{R}^4$  be written as a linear combination of the columns of the matrix  $A$  above? Do the columns of  $A$  span  $\mathbb{R}^4$ ?
20. Do the columns of  $B$  span  $\mathbb{R}^4$ ? Does the equation  $Bx = y$  have a solution for each  $y$  in  $\mathbb{R}^4$ ?

21. Let  $v_1 = \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix}$ ,  $v_2 = \begin{bmatrix} 0 \\ -1 \\ 0 \\ 1 \end{bmatrix}$ ,  $v_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ -1 \end{bmatrix}$ . Does

$\{v_1, v_2, v_3\}$  span  $\mathbb{R}^4$ ? Why or why not?

22. Let  $v_1 = \begin{bmatrix} 0 \\ 0 \\ -3 \end{bmatrix}$ ,  $v_2 = \begin{bmatrix} 0 \\ -3 \\ 9 \end{bmatrix}$ ,  $v_3 = \begin{bmatrix} 4 \\ -2 \\ -6 \end{bmatrix}$ . Does

$\{v_1, v_2, v_3\}$  span  $\mathbb{R}^3$ ? Why or why not?

In Exercises 23 and 24, mark each statement True or False. Justify each answer.

23. a. The equation  $Ax = b$  is referred to as a *vector equation*.  
 b. A vector  $b$  is a linear combination of the columns of a matrix  $A$  if and only if the equation  $Ax = b$  has at least one solution.  
 c. The equation  $Ax = b$  is consistent if the augmented matrix  $[A \ b]$  has a pivot position in every row.  
 d. The first entry in the product  $Ax$  is a sum of products.  
 e. If the columns of an  $m \times n$  matrix  $A$  span  $\mathbb{R}^m$ , then the equation  $Ax = b$  is consistent for each  $b$  in  $\mathbb{R}^m$ .  
 f. If  $A$  is an  $m \times n$  matrix and if the equation  $Ax = b$  is inconsistent for some  $b$  in  $\mathbb{R}^m$ , then  $A$  cannot have a pivot position in every row.

24. a. Every matrix equation  $Ax = b$  corresponds to a vector equation with the same solution set.  
 b. If the equation  $Ax = b$  is consistent, then  $b$  is in the set spanned by the columns of  $A$ .  
 c. Any linear combination of vectors can always be written in the form  $Ax$  for a suitable matrix  $A$  and vector  $x$ .  
 d. If the coefficient matrix  $A$  has a pivot position in every row, then the equation  $Ax = b$  is inconsistent.  
 e. The solution set of a linear system whose augmented matrix is  $[a_1 \ a_2 \ a_3 \ b]$  is the same as the solution set of  $Ax = b$ , if  $A = [a_1 \ a_2 \ a_3]$ .  
 f. If  $A$  is an  $m \times n$  matrix whose columns do not span  $\mathbb{R}^m$ , then the equation  $Ax = b$  is consistent for every  $b$  in  $\mathbb{R}^m$ .

25. Note that  $\begin{bmatrix} 4 & -3 & 1 \\ 5 & -2 & 5 \\ -6 & 2 & -3 \end{bmatrix} \begin{bmatrix} -3 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} -7 \\ -3 \\ 10 \end{bmatrix}$ . Use this

fact (and no row operations) to find scalars  $c_1, c_2, c_3$  such

$$\text{that } \begin{bmatrix} -7 \\ -3 \\ 10 \end{bmatrix} = c_1 \begin{bmatrix} 4 \\ 5 \\ -6 \end{bmatrix} + c_2 \begin{bmatrix} -3 \\ -2 \\ 2 \end{bmatrix} + c_3 \begin{bmatrix} 1 \\ 5 \\ -3 \end{bmatrix}.$$

26. Let  $u = \begin{bmatrix} 7 \\ 2 \\ 5 \end{bmatrix}$ ,  $v = \begin{bmatrix} 3 \\ 1 \\ 3 \end{bmatrix}$ , and  $w = \begin{bmatrix} 5 \\ 1 \\ 1 \end{bmatrix}$ . It can be shown that  $2u - 3v - w = 0$ . Use this fact (and no row operations) to find  $x_1$  and  $x_2$  that satisfy the equation

$$\begin{bmatrix} 7 & 3 \\ 2 & 1 \\ 5 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 5 \\ 1 \\ 1 \end{bmatrix}.$$

27. Rewrite the (numerical) matrix equation below in symbolic form as a vector equation, using symbols  $v_1, v_2, \dots$  for the vectors and  $c_1, c_2, \dots$  for scalars. Define what each symbol represents, using the data given in the matrix equation.

$$\begin{bmatrix} -3 & 5 & -4 & 9 & 7 \\ 5 & 8 & 1 & -2 & -4 \end{bmatrix} \begin{bmatrix} -3 \\ 1 \\ 2 \\ -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 11 \\ -11 \end{bmatrix}$$

28. Let  $q_1, q_2, q_3$ , and  $v$  represent vectors in  $\mathbb{R}^3$ , and let  $x_1, x_2$ , and  $x_3$  denote scalars. Write the following vector equation as a matrix equation. Identify any symbols you choose to use.

$$x_1 q_1 + x_2 q_2 + x_3 q_3 = v$$

29. Construct a  $3 \times 3$  matrix, not in echelon form, whose columns span  $\mathbb{R}^3$ . Show that the matrix you construct has the desired property.
30. Construct a  $3 \times 3$  matrix, not in echelon form, whose columns do not span  $\mathbb{R}^3$ . Show that the matrix you construct has the desired property.
31. Let  $A$  be a  $3 \times 2$  matrix. Explain why the equation  $Ax = b$  cannot be consistent for all  $b$  in  $\mathbb{R}^3$ . Generalize your argument to the case of an arbitrary  $A$  with more rows than columns.

# Homogeneous Systems

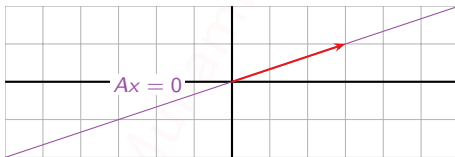
Example, continued

## Question

What is the solution set of  $Ax = 0$ , where

$$A = \begin{pmatrix} 1 & -3 \\ 2 & -6 \end{pmatrix}?$$

**Answer:**  $x = x_2 \begin{pmatrix} 3 \\ 1 \end{pmatrix}$  for any  $x_2$  in  $\mathbf{R}$ . The solution set is  $\text{Span}\left\{\begin{pmatrix} 3 \\ 1 \end{pmatrix}\right\}$ .



**Note:** one free variable means the solution set is a *line* in  $\mathbf{R}^2$  ( $2 = \#$  variables  $= \#$  columns).

# Homogeneous Systems

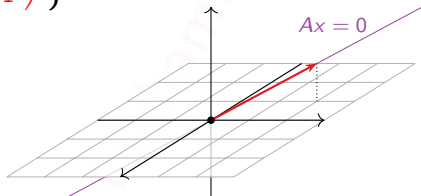
## Example

### Question

What is the solution set of  $Ax = 0$ , where

$$A = \begin{pmatrix} 1 & 3 & 1 \\ 2 & -1 & -5 \\ 1 & 0 & -2 \end{pmatrix}?$$

**Answer:**  $\text{Span} \left\{ \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \right\}.$



**Note:** one free variable means the solution set is a *line* in  $\mathbf{R}^3$  ( $3 = \#$  variables  $= \#$  columns).



# Homogeneous Systems

## Example

### Question

What is the solution set of  $Ax = 0$ , where

$$A = \begin{pmatrix} 1 & 2 & 0 & -1 \\ -2 & -3 & 4 & 5 \\ 2 & 4 & 0 & -2 \end{pmatrix}?$$

**Answer:**  $\text{Span} \left\{ \begin{pmatrix} 8 \\ -4 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 7 \\ -3 \\ 0 \\ 1 \end{pmatrix} \right\}.$

[not pictured here]

**Note:** two free variables means the solution set is a *plane* in  $\mathbf{R}^4$  ( $4 = \#$  variables =  $\#$  columns).

# Parametric Vector Form

## Homogeneous systems

Let  $A$  be an  $m \times n$  matrix. Suppose that the free variables in the homogeneous equation  $Ax = 0$  are  $x_i, x_j, x_k, \dots$

Then the solutions to  $Ax = 0$  can be written in the form

$$x = x_i v_i + x_j v_j + x_k v_k + \dots$$

for some vectors  $v_i, v_j, v_k, \dots$  in  $\mathbf{R}^n$ , and any scalars  $x_i, x_j, x_k, \dots$

The solution set is

$$\text{Span}\{v_i, v_j, v_k, \dots\}.$$

The **equation** above is called the **parametric vector form** of the solution.

## Poll

How many solutions can there be to a homogeneous system with more equations than variables?

- A. 0
- B. 1
- C.  $\infty$

The trivial solution is always a solution to a homogeneous system, so answer A is impossible.

This matrix has only one solution to  $Ax = 0$ :

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{pmatrix}$$

This matrix has infinitely many solutions to  $Ax = 0$ :

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 0 \\ 0 & 0 \end{pmatrix}$$

# Nonhomogeneous Systems

## Example

### Question

What is the solution set of  $Ax = b$ , where

$$A = \begin{pmatrix} 1 & -3 \\ 2 & -6 \end{pmatrix} \quad \text{and} \quad b = \begin{pmatrix} -3 \\ -6 \end{pmatrix}?$$

$$\left( \begin{array}{cc|c} 1 & -3 & -3 \\ 2 & -6 & -6 \end{array} \right) \xrightarrow{\text{row reduce}} \left( \begin{array}{cc|c} 1 & -3 & -3 \\ 0 & 0 & 0 \end{array} \right)$$

$$\xrightarrow{\text{equation}} x_1 - 3x_2 = -3$$

$$\xrightarrow{\text{parametric form}} \begin{cases} x_1 = 3x_2 - 3 \\ x_2 = x_2 + 0 \end{cases}$$

$$\xrightarrow{\text{parametric vector form}} x = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = x_2 \begin{pmatrix} 3 \\ 1 \end{pmatrix} + \begin{pmatrix} -3 \\ 0 \end{pmatrix}.$$

The only difference from the homogeneous case is the constant vector  $p = \begin{pmatrix} -3 \\ 0 \end{pmatrix}$ .

Note that  $p$  is itself a solution: take  $x_2 = 0$ .

# Nonhomogeneous Systems

Example, continued

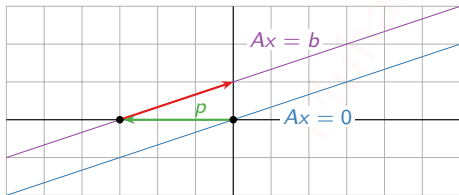
## Question

What is the solution set of  $Ax = b$ , where

$$A = \begin{pmatrix} 1 & -3 \\ 2 & -6 \end{pmatrix} \quad \text{and} \quad b = \begin{pmatrix} -3 \\ -6 \end{pmatrix}?$$

**Answer:**  $x = x_2 \begin{pmatrix} 3 \\ 1 \end{pmatrix} + \begin{pmatrix} -3 \\ 0 \end{pmatrix}$  for any  $x_2$  in  $\mathbf{R}$ .

This is a *translate* of  $\text{Span}\left\{\begin{pmatrix} 3 \\ 1 \end{pmatrix}\right\}$ : it is the parallel line through  $p = \begin{pmatrix} -3 \\ 0 \end{pmatrix}$ .



It can be written

$$\text{Span}\left\{\begin{pmatrix} 3 \\ 1 \end{pmatrix}\right\} + \begin{pmatrix} -3 \\ 0 \end{pmatrix}.$$

# Nonhomogeneous Systems

## Example

### Question

What is the solution set of  $Ax = b$ , where

$$A = \begin{pmatrix} 1 & 3 & 1 \\ 2 & -1 & -5 \\ 1 & 0 & -2 \end{pmatrix} \quad \text{and} \quad b = \begin{pmatrix} -5 \\ -3 \\ -2 \end{pmatrix} ?$$

$$\left( \begin{array}{ccc|c} 1 & 3 & 1 & -5 \\ 2 & -1 & -5 & -3 \\ 1 & 0 & -2 & -2 \end{array} \right) \xrightarrow{\text{row reduce}} \left( \begin{array}{ccc|c} 1 & 0 & -2 & -2 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

$$\xrightarrow{\text{equations}} \begin{cases} x_1 - 2x_3 = -2 \\ x_2 + x_3 = -1 \end{cases}$$

$$\xrightarrow{\text{parametric form}} \begin{cases} x_1 = 2x_3 - 2 \\ x_2 = -x_3 - 1 \\ x_3 = x_3 \end{cases}$$

$$\xrightarrow{\text{parametric vector form}} x = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = x_3 \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} + \begin{pmatrix} -2 \\ -1 \\ 0 \end{pmatrix}.$$

# Nonhomogeneous Systems

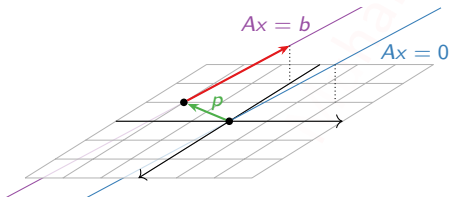
Example, continued

## Question

What is the solution set of  $Ax = b$ , where

$$A = \begin{pmatrix} 1 & 3 & 1 \\ 2 & -1 & -5 \\ 1 & 0 & -2 \end{pmatrix} \quad \text{and} \quad b = \begin{pmatrix} -5 \\ -3 \\ -2 \end{pmatrix}?$$

**Answer:**  $\text{Span} \left\{ \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \right\} + \begin{pmatrix} -2 \\ -1 \\ 0 \end{pmatrix}.$



The solution set is a *translate* of

$$\text{Span} \left\{ \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} \right\} :$$

it is the parallel line through

$$p = \begin{pmatrix} -2 \\ -1 \\ 0 \end{pmatrix}.$$

## Homogeneous vs. Nonhomogeneous Systems

### Key Observation

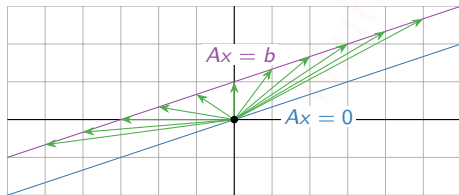
The set of solutions to  $Ax = b$ , if it is nonempty, is obtained by taking one **specific** or **particular solution**  $p$  to  $Ax = b$ , and adding all solutions to  $Ax = 0$ .

**Why?** If  $Ap = b$  and  $Ax = 0$ , then

$$A(p + x) = Ap + Ax = b + 0 = b,$$

so  $p + x$  is also a solution to  $Ax = b$ .

We know the solution set of  $Ax = 0$  is a span. So the solution set of  $Ax = b$  is a *translate* of a span: it is *parallel* to a span. (Or it is empty.)



This works for *any* specific solution  $p$ : it doesn't have to be the one produced by finding the parametric vector form and setting the free variables all to zero, as we did before.

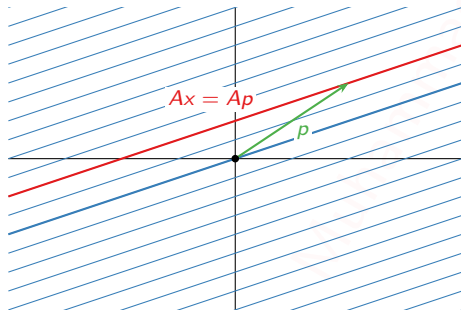


# Homogeneous vs. Nonhomogeneous Systems

## Varying $b$

If we understand the solution set of  $Ax = 0$ , then we understand the solution set of  $Ax = b$  for all  $b$ : they are all translates (or empty).

For instance, if  $A = \begin{pmatrix} 1 & -3 \\ 2 & -6 \end{pmatrix}$ , then the solution sets for varying  $b$  look like this:



Which  $b$  gives the solution set  $Ax = b$  in red in the picture?

Choose  $p$  on the red line, and set  $b = Ap$ . Then  $p$  is a specific solution to  $Ax = b$ , so the solution set of  $Ax = b$  is the red line.

Note the cool optical illusion!

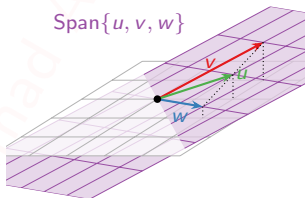
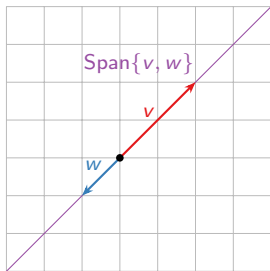
For a matrix equation  $Ax = b$ , you now know how to find which  $b$ 's are possible, and what the solution set looks like for all  $b$ , both using spans.

# Section 1.7

## Linear Independence

## Motivation

Sometimes the span of a set of vectors is “smaller” than you expect from the number of vectors.



This can mean many things. For example, it can mean you're using too many vectors to write your solution set.

Notice in each case that one vector in the set is already in the span of the others—so it doesn't make the span bigger.

Today we will formalize this idea in the concept of *linear (in)dependence*.

# Linear Independence

## Definition

A set of vectors  $\{v_1, v_2, \dots, v_p\}$  in  $\mathbf{R}^n$  is **linearly independent** if the vector equation

$$x_1 v_1 + x_2 v_2 + \dots + x_p v_p = 0$$

has only the trivial solution  $x_1 = x_2 = \dots = x_p = 0$ . The set  $\{v_1, v_2, \dots, v_p\}$  is **linearly dependent** otherwise.

In other words,  $\{v_1, v_2, \dots, v_p\}$  is linearly dependent if there exist numbers  $x_1, x_2, \dots, x_p$ , not all equal to zero, such that

$$x_1 v_1 + x_2 v_2 + \dots + x_p v_p = 0.$$

This is called a **linear dependence relation**.

Like span, linear (in)dependence is another one of those big vocabulary words that you absolutely need to learn. Much of the rest of the course will be built on these concepts, and you need to know exactly what they mean in order to be able to answer questions on quizzes and exams (and solve real-world problems later on).

# Linear Independence

## Definition

A set of vectors  $\{v_1, v_2, \dots, v_p\}$  in  $\mathbf{R}^n$  is **linearly independent** if the vector equation

$$x_1 v_1 + x_2 v_2 + \dots + x_p v_p = 0$$

has only the trivial solution  $x_1 = x_2 = \dots = x_p = 0$ . The set  $\{v_1, v_2, \dots, v_p\}$  is **linearly dependent** otherwise.

Note that linear (in)dependence is a notion that applies to a *collection of vectors*, not to a single vector, or to one vector in the presence of some others.

## Checking Linear Independence

**Question:** Is  $\left\{ \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}, \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix} \right\}$  linearly independent?

Equivalently, does the (homogeneous) the vector equation

$$x \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + y \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} + z \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

have a nontrivial solution?

How do we solve this kind of vector equation?

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & 1 \\ 1 & 2 & 4 \end{pmatrix} \xrightarrow{\text{row reduce}} \begin{pmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$$

So  $x = -2z$  and  $y = -z$ . So the vectors are linearly dependent, and an equation of linear dependence is (taking  $z = 1$ )

$$-2 \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} + \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

## Checking Linear Independence

**Question:** Is  $\left\{ \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}, \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix} \right\}$  linearly independent?

Equivalently, does the (homogeneous) the vector equation

$$x \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix} + y \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix} + z \begin{pmatrix} 3 \\ 1 \\ 4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

have a nontrivial solution?

$$\begin{pmatrix} 1 & 1 & 3 \\ 1 & -1 & 1 \\ 0 & 2 & 4 \end{pmatrix} \xrightarrow{\text{row reduce}} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

The trivial solution  $\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$  is the unique solution. So the vectors are linearly *independent*.

## Linear Independence and Matrix Columns

In general,  $\{v_1, v_2, \dots, v_p\}$  is linearly independent if and only if the vector equation

$$x_1 v_1 + x_2 v_2 + \dots + x_p v_p = 0$$

has only the trivial solution, if and only if the matrix equation

$$Ax = 0$$

has only the trivial solution, where  $A$  is the matrix with columns  $v_1, v_2, \dots, v_p$ :

$$A = \begin{pmatrix} | & | & \cdots & | \\ v_1 & v_2 & \cdots & v_p \\ | & | & \cdots & | \end{pmatrix}.$$

This is true if and only if the matrix  $A$  has a pivot in each column.

### Important

- ▶ The vectors  $v_1, v_2, \dots, v_p$  are linearly independent if and only if the matrix with columns  $v_1, v_2, \dots, v_p$  has a pivot in each column.
- ▶ Solving the matrix equation  $Ax = 0$  will either verify that the columns  $v_1, v_2, \dots, v_p$  of  $A$  are linearly independent, or will produce a linear dependence relation.



# Linear Independence

## Criterion

Suppose that one of the vectors  $\{v_1, v_2, \dots, v_p\}$  is a linear combination of the other ones (that is, it is in the span of the other ones):

$$v_3 = 2v_1 - \frac{1}{2}v_2 + 6v_4$$

Then the vectors are linearly dependent:

$$2v_1 - \frac{1}{2}v_2 - v_3 + 6v_4 = 0.$$

Conversely, if the vectors are linearly dependent

$$2v_1 - \frac{1}{2}v_2 + 6v_4 = 0.$$

then one vector is a linear combination of (in the span of) the other ones:

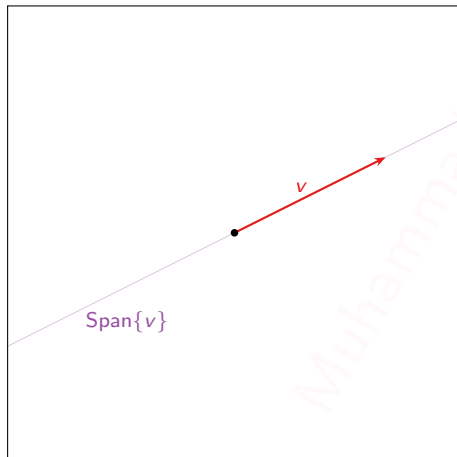
$$v_2 = 4v_1 + 12v_4.$$

## Theorem

A set of vectors  $\{v_1, v_2, \dots, v_p\}$  is linearly dependent if and only if one of the vectors is in the span of the other ones.

# Linear Independence

Pictures in  $\mathbb{R}^2$



In this picture

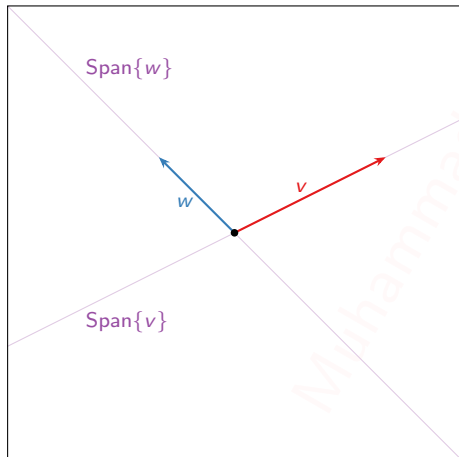
---

One vector  $\{v\}$ :

Linearly independent if  $v \neq 0$ .

# Linear Independence

Pictures in  $\mathbb{R}^2$



In this picture

---

One vector  $\{v\}$ :

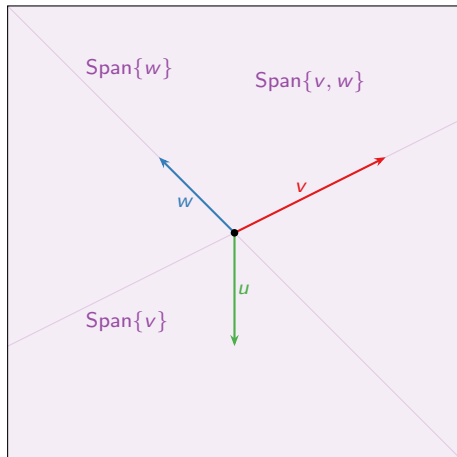
Linearly independent if  $v \neq 0$ .

Two vectors  $\{v, w\}$ :

Linearly independent: neither is in the span of the other.

# Linear Independence

Pictures in  $\mathbb{R}^2$



In this picture

**One vector  $\{v\}$ :**

Linearly independent if  $v \neq 0$ .

**Two vectors  $\{v, w\}$ :**

Linearly independent: neither is in the span of the other.

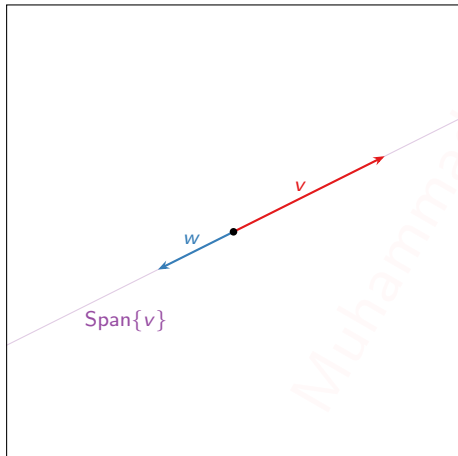
**Three vectors  $\{v, w, u\}$ :**

Linearly dependent:  $u$  is in  $\text{Span}\{v, w\}$ .

Also  $v$  is in  $\text{Span}\{u, w\}$  and  $w$  is in  $\text{Span}\{u, v\}$ .

# Linear Independence

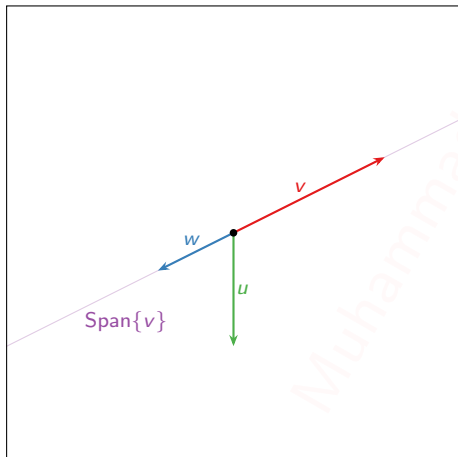
Pictures in  $\mathbb{R}^2$



Two collinear vectors  $\{v, w\}$ :  
Linearly dependent:  $w$  is in  $\text{Span}\{v\}$  (and vice-versa).

# Linear Independence

Pictures in  $\mathbb{R}^2$



**Two collinear vectors  $\{v, w\}$ :**

Linearly dependent:  $w$  is in  $\text{Span}\{v\}$  (and vice-versa).

**Observe:** Two vectors are linearly dependent if and only if they are *collinear*.

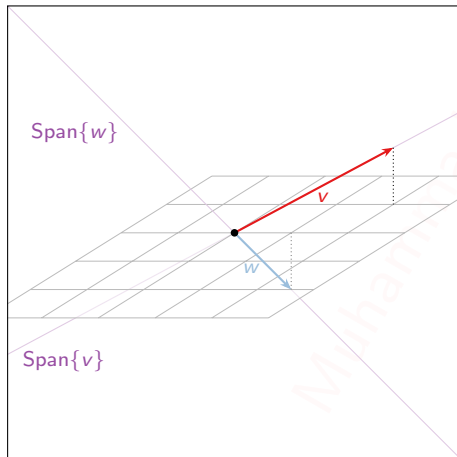
**Three vectors  $\{v, w, u\}$ :**

Linearly dependent:  $w$  is in  $\text{Span}\{v\}$  (and vice-versa).

**Observe:** If a set of vectors is linearly dependent, then so is any larger set of vectors!

# Linear Independence

Pictures in  $\mathbb{R}^3$



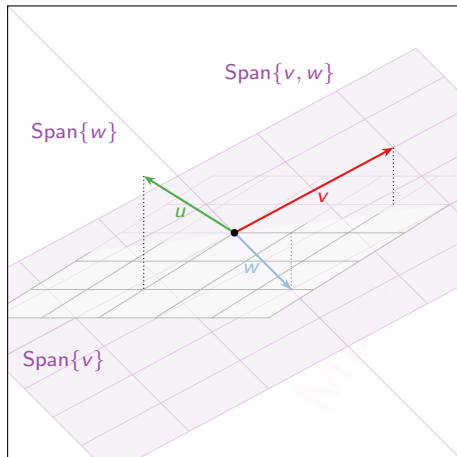
In this picture

Two vectors  $\{v, w\}$ :

Linearly independent: neither is in the span of the other.

# Linear Independence

Pictures in  $\mathbb{R}^3$



In this picture

Two vectors  $\{v, w\}$ :

Linearly independent: neither is in the span of the other.

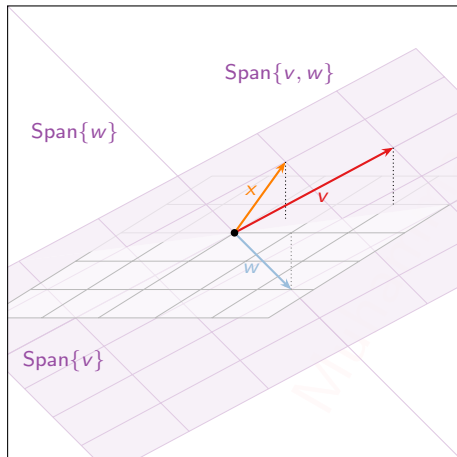
Three vectors  $\{v, w, u\}$ :

Linearly independent: no one is in the span of the other two.



# Linear Independence

Pictures in  $\mathbb{R}^3$



In this picture

Two vectors  $\{v, w\}$ :

Linearly independent: neither is in the span of the other.

Three vectors  $\{v, w, x\}$ :

Linearly dependent:  $x$  is in  $\text{Span}\{v, w\}$ .

## Poll

Are there four vectors  $u, v, w, x$  in  $\mathbf{R}^3$  which are linearly dependent, but such that  $u$  is *not* a linear combination of  $v, w, x$ ? If so, draw a picture; if not, give an argument.

**Yes:** actually the pictures on the previous slides provide such an example.

Linear dependence of  $\{v_1, \dots, v_p\}$  means *some*  $v_i$  is a linear combination of the others, not *any*.

# Linear Independence

## Stronger criterion

### Theorem

A set of vectors  $\{v_1, v_2, \dots, v_p\}$  is linearly dependent if and only if one of the vectors is in the span of the other ones.

Take the largest  $j$  such that  $v_j$  is in the span of the others. Then  $v_j$  is in the span of  $v_1, v_2, \dots, v_{j-1}$ . Why? If not ( $j = 3$ ):

$$v_3 = 2v_1 - \frac{1}{2}v_2 + 6v_4$$

Rearrange:

$$v_4 = -\frac{1}{6} \left( 2v_1 - \frac{1}{2}v_2 - v_3 \right)$$

so  $v_4$  works as well, but  $v_3$  was supposed to be the last one that was in the span of the others.

### Better Theorem

A set of vectors  $\{v_1, v_2, \dots, v_p\}$  is linearly dependent if and only if there is some  $j$  such that  $v_j$  is in  $\text{Span}\{v_1, v_2, \dots, v_{j-1}\}$ .

# Linear Independence

## Increasing span criterion

### Theorem

A set of vectors  $\{v_1, v_2, \dots, v_p\}$  is linearly dependent if and only if there is some  $j$  such that  $v_j$  is in  $\text{Span}\{v_1, v_2, \dots, v_{j-1}\}$ .

Equivalently,  $\{v_1, v_2, \dots, v_p\}$  is linearly *independent* if for every  $j$ , the vector  $v_j$  is not in  $\text{Span}\{v_1, v_2, \dots, v_{j-1}\}$ .

This means  $\text{Span}\{v_1, v_2, \dots, v_j\}$  is *bigger* than  $\text{Span}\{v_1, v_2, \dots, v_{j-1}\}$ .

### Theorem

A set of vectors  $\{v_1, v_2, \dots, v_p\}$  is linearly independent if and only if, for every  $j$ , the span of  $v_1, v_2, \dots, v_j$  is strictly larger than the span of  $v_1, v_2, \dots, v_{j-1}$ .

#### Translation

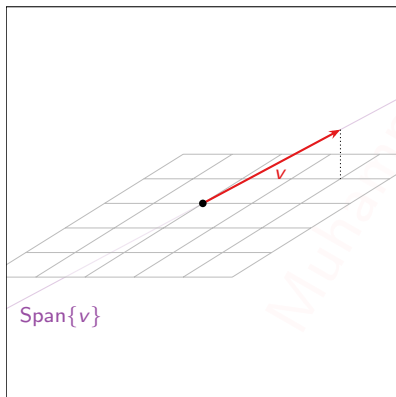
A set of vectors is linearly independent if and only if, every time you add another vector to the set, the span gets bigger.

# Linear Independence

Increasing span criterion: pictures

## Theorem

A set of vectors  $\{v_1, v_2, \dots, v_p\}$  is linearly independent if and only if, for every  $j$ , the span of  $v_1, v_2, \dots, v_j$  is strictly larger than the span of  $v_1, v_2, \dots, v_{j-1}$ .



One vector  $\{v\}$ :

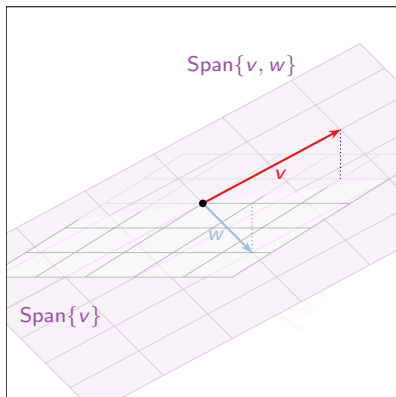
Linearly independent: span got bigger (than  $\{0\}$ ).

# Linear Independence

Increasing span criterion: pictures

## Theorem

A set of vectors  $\{v_1, v_2, \dots, v_p\}$  is linearly independent if and only if, for every  $j$ , the span of  $v_1, v_2, \dots, v_j$  is strictly larger than the span of  $v_1, v_2, \dots, v_{j-1}$ .



One vector  $\{v\}$ :

Linearly independent: span got bigger (than  $\{0\}$ ).

Two vectors  $\{v, w\}$ :

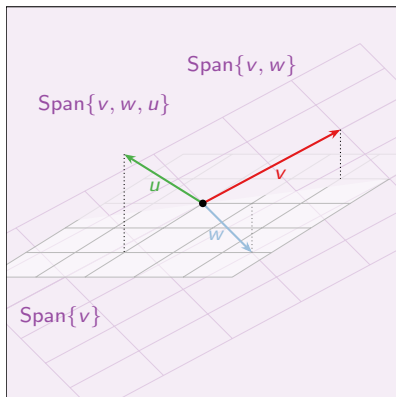
Linearly independent: span got bigger.

# Linear Independence

Increasing span criterion: pictures

## Theorem

A set of vectors  $\{v_1, v_2, \dots, v_p\}$  is linearly independent if and only if, for every  $j$ , the span of  $v_1, v_2, \dots, v_j$  is strictly larger than the span of  $v_1, v_2, \dots, v_{j-1}$ .



One vector  $\{v\}$ :

Linearly independent: span got bigger (than  $\{0\}$ ).

Two vectors  $\{v, w\}$ :

Linearly independent: span got bigger.

Three vectors  $\{v, w, u\}$ :

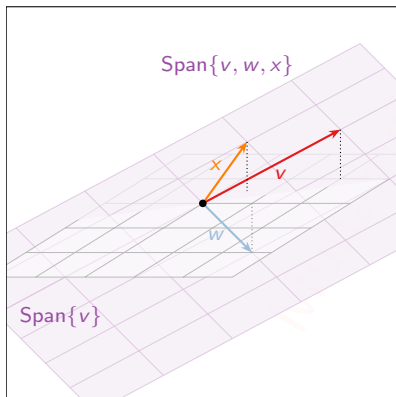
Linearly independent: span got bigger.

# Linear Independence

Increasing span criterion: pictures

## Theorem

A set of vectors  $\{v_1, v_2, \dots, v_p\}$  is linearly independent if and only if, for every  $j$ , the span of  $v_1, v_2, \dots, v_j$  is strictly larger than the span of  $v_1, v_2, \dots, v_{j-1}$ .



One vector  $\{v\}$ :

Linearly independent: span got bigger (than  $\{0\}$ ).

Two vectors  $\{v, w\}$ :

Linearly independent: span got bigger.

Three vectors  $\{v, w, x\}$ :

Linearly dependent: span didn't get bigger.



# Linear Independence

## Two more facts

**Fact 1:** Say  $v_1, v_2, \dots, v_n$  are in  $\mathbf{R}^m$ . If  $n > m$  then  $\{v_1, v_2, \dots, v_n\}$  is linearly dependent: the matrix

$$A = \begin{pmatrix} | & | & \cdots & | \\ v_1 & v_2 & \cdots & v_n \\ | & | & \cdots & | \end{pmatrix}.$$

cannot have a pivot in each column (it is too wide).

This says you can't have 4 linearly independent vectors in  $\mathbf{R}^3$ , for instance.

A wide matrix can't have linearly independent columns.

**Fact 2:** If one of  $v_1, v_2, \dots, v_n$  is zero, then  $\{v_1, v_2, \dots, v_n\}$  is linearly dependent. For instance, if  $v_1 = 0$ , then

$$1 \cdot v_1 + 0 \cdot v_2 + 0 \cdot v_3 + \cdots + 0 \cdot v_n = 0$$

is a linear dependence relation.

A set containing the zero vector is linearly dependent.

- ▶ The columns of a matrix  $A$  are linearly independent if the equation  $Ax = 0$  has the trivial solution.
- ▶ If  $S$  is a linearly dependent set, then each vector is a linear combination of the other vectors in  $S$ .
- ▶ The columns of any  $4 \times 5$  matrix are linearly dependent. If  $x$  and  $y$  are linearly independent, and if  $\{x; y; z\}$  is linearly dependent, then  $z$  is in  $\text{Span}\{x; y\}$
- ▶ If  $\{v_1, \dots, v_5\}$  are in  $R^5$  and  $v_3 = 0$ , then  $\{v_1, \dots, v_5\}$  is linearly dependent.
- ▶ Suppose  $A$  is an  $m \times n$  matrix with the property that for all  $b$  in  $R^m$  the equation  $Ax = b$  has at most one solution. Explain why the columns of  $A$  must be linearly independent.

- (a). Find the value(s) of  $h$  for which the vectors are linearly dependent. Is

$$\left\{ \begin{pmatrix} 2 \\ -2 \\ 4 \end{pmatrix}, \begin{pmatrix} 4 \\ -6 \\ 7 \end{pmatrix}, \begin{pmatrix} -2 \\ 2 \\ h \end{pmatrix} \right\} \text{ linearly independent?}$$

- (b). (i) For what values of  $h$  is  $v_3$  in  $\text{Span}\{v_1; v_2\}$ , and (ii) for what values of  $h$  is  $\{v_1; v_2; v_3\}$  linearly dependent?

$$\left\{ \begin{pmatrix} 1 \\ -3 \\ 2 \end{pmatrix}, \begin{pmatrix} -3 \\ 9 \\ -6 \end{pmatrix}, \begin{pmatrix} 5 \\ -7 \\ h \end{pmatrix} \right\}$$

- (c). Determine by inspection whether the vectors are linearly independent

(i).  $\left\{ \begin{pmatrix} 5 \\ 8 \end{pmatrix}, \begin{pmatrix} 2 \\ 1 \end{pmatrix}, \begin{pmatrix} 3 \\ 7 \end{pmatrix} \right\}$

(ii).  $\left\{ \begin{pmatrix} 5 \\ 3 \\ -1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} -7 \\ 2 \\ 4 \end{pmatrix} \right\}$

(iii).  $\left\{ \begin{pmatrix} 3 \\ 4 \end{pmatrix}, \begin{pmatrix} -1 \\ 5 \end{pmatrix}, \begin{pmatrix} 3 \\ 5 \end{pmatrix}, \begin{pmatrix} 7 \\ 1 \end{pmatrix} \right\}$